

# Capacity Region of the Binary Symmetric Z-Interference Channel: A Direct Derivation

[Author list TBD]

## Abstract

We present a compact, self-contained derivation of the capacity region of the two-user binary symmetric Z-interference channel  $Y_1 = X_1 \oplus X_2 \oplus N_1$ ,  $Y_2 = X_2 \oplus N_2$ , where the noises are independent and  $N_i \sim \text{Bern}(p_i)$  with  $p_i \in (0, \frac{1}{2})$ . If  $p_1 \leq p_2$ , the strong-interference condition holds and the capacity region is  $\{(R_1, R_2) \geq 0 : R_1 + R_2 \leq 1 - h(p_1), R_2 \leq 1 - h(p_2)\}$ . If  $p_1 > p_2$ , the region equals the degraded binary symmetric broadcast-channel region with crossover probabilities  $(p_1, p_2)$ . Every boundary point in this weak regime is achieved by independent point-to-point coding with a uniform input for user 1, a suitably biased input for user 2, and treating interference as noise at receiver 1; neither rate splitting nor time sharing is needed to trace the boundary. The unique sum-capacity rate pair is  $(0, 1 - h(p_2))$ . The characterization is classical in substance, following from strong-interference theory and degraded additive-interference results; the purpose of this note is to assemble the two regimes in one explicit formula and give a direct weak-regime converse based on Mrs. Gerber's Lemma.

## I. Introduction

The two-user interference channel is a canonical model for communication in the presence of unintended signals. Its one-sided, or Z-interference, specialization isolates a single interference link while retaining the basic choices of interference management: decoding the interference, partially decoding it, or treating it as noise. Even for the Gaussian Z-interference channel, the full weak-interference capacity region remains unresolved [9], [18], [19].

This note considers the binary additive model

$$Y_1 = X_1 \oplus X_2 \oplus N_1, \quad Y_2 = X_2 \oplus N_2, \quad (1)$$

where  $N_1 \sim \text{Bern}(p_1)$  and  $N_2 \sim \text{Bern}(p_2)$  are independent of each other and of the inputs, and  $p_1, p_2 \in (0, \frac{1}{2})$ . Receiver  $i$  decodes only message  $i$ , and the encoders are independent.

The value of the result is primarily expository and structural rather than a new general capacity theorem. First, it places the strong and weak parameter regimes in a single closed-form statement. Second, in the weak regime it reduces the converse to a direct application of Mrs. Gerber’s Lemma [1], without introducing an auxiliary random variable or first invoking broadcast-channel equivalence. Third, it shows directly that every boundary point is attained by a single product input distribution: receiver 1 treats user 2’s signal as noise, while user 2 adjusts its input bias. Finally, it makes explicit the precise correspondence with the degraded binary symmetric broadcast channel [2], and records a structural feature of independent interest: this channel has no very-strong-interference regime (Section V).

#### A. Related work and scope of the contribution

The strong-interference capacity region was developed by Carleial [3] and Sato [4], [7], and established for the discrete memoryless interference channel by Costa and El Gamal [10]; see also [16, Ch. 6] and, for deterministic/additive structure, El Gamal and Costa [8]. Part (i) of the main theorem is a direct specialization of this theory.

For degraded additive interference channels, Benzel [5] gave a single-letter capacity characterization and identified the corresponding degraded-broadcast-channel region. Liu and Ulukus [11] extended the degraded-interference class and proved that encoder cooperation does not enlarge its capacity region. Liu and Goldsmith [15] later identified a class of Z-interference channels that includes additive modulo channels such as (1), and gave a single-letter capacity characterization for the relevant subclass. The binary symmetric broadcast boundary itself follows from Mrs. Gerber’s Lemma and Wyner’s application of it to the degraded broadcast channel [1], [2].

Consequently, the capacity region stated here should be viewed as an explicit specialization and synthesis of classical results, not as a claim of priority for the region itself. The contribution is the all-parameter closed form, a short direct converse for the weak regime, an equally direct achievability calculation, and the strict identification of the unique sum-capacity rate pair. The general Han–Kobayashi construction is relevant background [6], but no partial message splitting is required for this particular channel.

## II. Model and Main Result

Let

$$h(x) = -x \log_2 x - (1 - x) \log_2(1 - x), \quad x \in [0, 1],$$

be the binary entropy function, let  $h^{-1} : [0, 1] \rightarrow [0, \frac{1}{2}]$  denote its inverse on  $[0, \frac{1}{2}]$ , and define binary convolution by

$$a \star b = a(1 - b) + b(1 - a).$$

Rates are measured in bits per channel use, and achievability is defined with vanishing average error probability. The restriction  $p_i \in (0, \frac{1}{2})$  excludes only degenerate cases:  $p_i = 0$  or  $p_i = \frac{1}{2}$  yield noiseless or useless links, and the corresponding regions follow from the theorem by continuity.

Theorem 1: The capacity region of the binary symmetric Z-interference channel (1) is as follows.

(i) If  $p_1 \leq p_2$ , then

$$\mathcal{C} = \{(R_1, R_2) \geq 0 : R_1 + R_2 \leq 1 - h(p_1), \quad R_2 \leq 1 - h(p_2)\}.$$

(ii) If  $p_1 > p_2$ , define

$$q = \frac{p_1 - p_2}{1 - 2p_2} \in (0, \frac{1}{2}), \quad p_1 = p_2 \star q.$$

Then

$$\mathcal{C} = \left\{ (R_1, R_2) \geq 0 : \begin{array}{l} R_2 \leq 1 - h(p_2), \\ R_1 \leq 1 - h(q \star h^{-1}(h(p_2) + R_2)) \end{array} \right\}.$$

In this regime the sum capacity is  $1 - h(p_2)$ , and the unique sum-capacity rate pair is  $(0, 1 - h(p_2))$ .

The weak-regime formula is the degraded binary symmetric broadcast-channel capacity region with crossover probabilities  $(p_1, p_2)$  after matching the weak-receiver rate with  $R_1$  and the strong-receiver private rate with  $R_2$  [2]. The two regions are shown in Fig. 1.

Remark 1 (Continuity across the regimes): At  $p_1 = p_2$  the two parts agree:  $q = 0$  makes the weak-regime constraint  $R_1 \leq 1 - h(h^{-1}(h(p_2) + R_2)) = 1 - h(p_2) - R_2$ , i.e. exactly the sum constraint of part (i) with  $1 - h(p_1) = 1 - h(p_2)$ .

### III. Strong-Interference Regime

a) Converse: Suppose  $p_1 \leq p_2$ . Then  $p_2 = p_1 \star p_0$  for some  $p_0 \in [0, \frac{1}{2})$ , so we may couple  $N_2^n = N_1^n \oplus N_0^n$ , with  $N_0^n$  i.i.d. Bern( $p_0$ ) and independent of all other variables. This coupling preserves the marginal channel from  $X_2^n$  to  $Y_2^n$  and gives

$$I(X_2^n; Y_2^n) \leq I(X_2^n; X_2^n \oplus N_1^n) = I(X_2^n; Y_1^n | X_1^n),$$

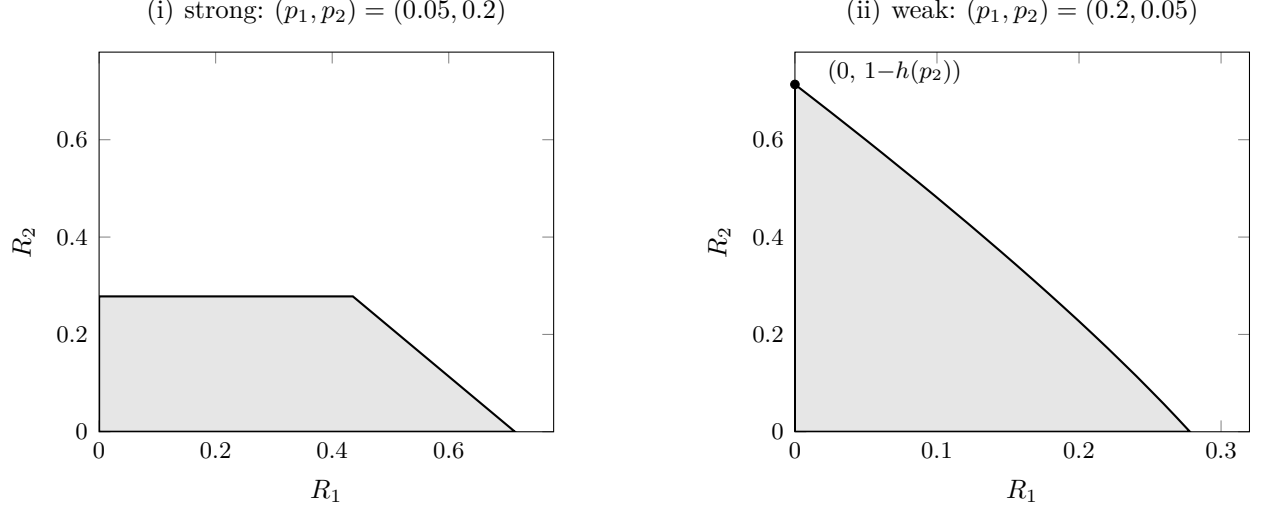


Fig. 1. Capacity regions of the BS-ZIC in the two regimes. Left: the strong-interference polytope of Theorem 1(i). Right: the weak-interference region of Theorem 1(ii); the marked corner is the unique sum-capacity rate pair.

where the equality uses independence of the encoders and the fact that, conditioned on  $X_1^n$ , adding  $X_1^n$  is a bijection.

By Fano's inequality and data processing, for some  $\epsilon_n \rightarrow 0$ ,

$$\begin{aligned}
 n(R_1 + R_2 - \epsilon_n) &\leq I(X_1^n; Y_1^n) + I(X_2^n; Y_2^n) \\
 &\leq I(X_1^n; Y_1^n) + I(X_2^n; Y_1^n \mid X_1^n) \\
 &= I(X_1^n, X_2^n; Y_1^n) \\
 &\leq n(1 - h(p_1)).
 \end{aligned}$$

The point-to-point channel to receiver 2 also gives  $R_2 \leq 1 - h(p_2)$ .

b) Achievability: Choose independent uniform inputs. Receiver 1 jointly decodes both messages on the binary additive multiple-access channel. Its achievable region is

$$R_1 \leq 1 - h(p_1), \quad R_2 \leq 1 - h(p_1), \quad R_1 + R_2 \leq 1 - h(p_1);$$

note that all three constraints share the same right-hand side, so receiver 1's region is a simplex rather than the usual pentagon: the modulo-2 sum collapses both codewords into a single binary observation. Receiver 2 requires  $R_2 \leq 1 - h(p_2)$ . Since  $p_1 \leq p_2$  implies  $1 - h(p_2) \leq 1 - h(p_1)$ , the resulting achievable region matches the converse. ■

#### IV. Weak-Interference Regime

a) Converse: Suppose  $p_1 > p_2$  and write  $p_1 = p_2 \star q$ , where  $q = (p_1 - p_2)/(1 - 2p_2)$ . Couple  $N_1^n = N_2^n \oplus Q^n$ , where  $Q^n$  is i.i.d. Bern( $q$ ) and independent of all other variables. Let

$$V^n = X_2^n \oplus N_2^n,$$

so  $V^n$  has the same distribution as  $Y_2^n$ . Fano's inequality at receiver 2 gives

$$\frac{1}{n}H(V^n) \geq h(p_2) + R_2 - \epsilon_n. \quad (2)$$

Define

$$f_q(t) = h(q \star h^{-1}(t)), \quad 0 \leq t \leq 1.$$

Mrs. Gerber's Lemma [1] states that

$$\frac{1}{n}H(V^n \oplus Q^n) \geq f_q\left(\frac{1}{n}H(V^n)\right).$$

The function  $f_q$  is increasing. Hence, using (2), Fano's inequality at receiver 1, and independence of the encoders,

$$\begin{aligned} R_1 - \epsilon_n &\leq \frac{1}{n}I(X_1^n; Y_1^n) \\ &\leq 1 - \frac{1}{n}H(Y_1^n | X_1^n) \\ &= 1 - \frac{1}{n}H(V^n \oplus Q^n) \\ &\leq 1 - f_q(h(p_2) + R_2 - \epsilon_n). \end{aligned}$$

For sufficiently large  $n$  the argument lies in  $[0, 1]$ ; endpoint cases follow by continuity. Letting  $n \rightarrow \infty$  proves the outer bound in Theorem 1(ii).

b) Achievability: Fix  $\pi \in [0, \frac{1}{2}]$ . Let user 2 employ an i.i.d. or constant-composition Bern( $\pi$ ) random code and let user 1 use a uniform random code. Receiver 2 achieves any rate below

$$I(X_2; Y_2) = h(\pi \star p_2) - h(p_2).$$

Receiver 1 treats  $X_2$  as noise. Since  $X_2 \oplus N_1 \sim \text{Bern}(\pi \star p_1)$ , it achieves any rate below

$$I(X_1; Y_1) = 1 - h(\pi \star p_1).$$

Set  $s = \pi \star p_2$ . Then  $h(s) = h(p_2) + R_2$  on the boundary, and

$$\pi \star p_1 = \pi \star (p_2 \star q) = (\pi \star p_2) \star q = s \star q.$$

Therefore

$$R_1 = 1 - h(q \star h^{-1}(h(p_2) + R_2)).$$

As  $\pi$  ranges over  $[0, \frac{1}{2}]$ ,  $s$  ranges over  $[p_2, \frac{1}{2}]$ , so this construction traces the entire outer-boundary curve. Rates below the boundary are obtained by coding below the two mutual informations; no time sharing is needed to trace the boundary. Moreover, none is needed anywhere:  $f_q$  is convex [1], so the boundary curve  $R_1 = 1 - f_q(h(p_2) + R_2)$  is concave in  $R_2$  and the region is convex as stated.

c) Unique sum-capacity rate pair: Along the boundary, parameterized by  $s \in [p_2, \frac{1}{2}]$ ,

$$R_1 + R_2 = 1 - h(p_2) + h(s) - h(q \star s).$$

If  $s < \frac{1}{2}$  and  $q > 0$ , then  $s < q \star s < \frac{1}{2}$ , and therefore  $h(s) < h(q \star s)$ . Thus the sum is strictly smaller than  $1 - h(p_2)$  unless  $s = \frac{1}{2}$ . Equality occurs only at  $(R_1, R_2) = (0, 1 - h(p_2))$ . ■

## V. Discussion

a) No very-strong-interference regime: In the Gaussian and general discrete interference channels, sufficiently strong interference eventually renders the sum constraint inactive and the capacity region becomes the interference-free rectangle. That never happens here: both messages must be extracted from the single binary observation  $Y_1$ , so  $I(X_1^n, X_2^n; Y_1^n) \leq n(1 - h(p_1))$  caps the sum rate for every  $p_1 \leq p_2$ , and the interfered user pays a rate penalty no matter how favorable its own channel. The modulo-2 collapse of the multiple-access geometry (the simplex noted above) is the mechanism.

b) Broadcast-channel correspondence: The weak-regime formula matches the degraded binary symmetric broadcast channel. Under the standard superposition parameterization, user 1's uniform codeword corresponds to the cloud-center variable carrying the weak-receiver message, while user 2's biased codeword corresponds to the binary satellite increment carrying the strong-receiver private message. This is a capacity-region equivalence, not a literal equality of channel laws: receiver 2 in (1) observes the satellite increment directly, whereas the strong broadcast receiver would first decode the cloud center and then recover the satellite layer. Benzel's degraded additive-interference result [5] and the Z-channel formulation of Liu and Goldsmith [15] formalize this equivalence.

c) A decode-all/TIN dichotomy: When  $p_1 \leq p_2$ , receiver 1 can decode user 2's message at least as reliably as receiver 2 can, once  $X_1^n$  is given; full interference decoding is optimal. When  $p_1 > p_2$ , the entire boundary is achieved by treating interference as noise with an optimized input bias. Thus the Han–Kobayashi partial-decoding mechanism [6] is unnecessary for this channel, although it remains fundamental for general interference channels. This exact discrete example complements broader Gaussian TIN-optimality results [17].

d) Binary versus Gaussian models: Mrs. Gerber's Lemma gives the exact entropy tradeoff produced by a binary symmetric degradation, which is why the weak-regime converse closes in a single step. No comparably tight extremal relation is known for the weak Gaussian Z-interference geometry: there, the sum capacity in the noisy-interference regime was settled by [12], [14], [13] (with TIN sum-optimal), while the full capacity region remains open, with recent progress refining inner and outer bounds [9], [18], [19]. In the binary symmetric case the entire region follows in a few lines.

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